# RoboServe: Design and Implementation of a Robotic Agent for Serving Food at Restaurants

Jul Jalal Al-Mamur Sayor
IoT and Robotics Engineering
Bangabandhu Sheikh Mujibur Rahman Digital University,Bangladesh
Dhaka, Bangladesh
1901029@iot.bdu.ac.bd

Nishat Tasnim Shishir
IoT and Robotics Engineering
Bangabandhu Sheikh Mujibur Rahman Digital University,Bangladesh
Dhaka, Bangladesh
1901030@iot.bdu.ac.bd

Bitta Boibhov Barmon
IoT and Robotics Engineering
Bangabandhu Sheikh Mujibur Rahman Digital University,Bangladesh
Dhaka, Bangladesh
1901050@iot.bdu.ac.bd

Sumon Ahemed
IoT and Robotics Engineering
Bangabandhu Sheikh Mujibur Rahman Digital university,Bangladesh
Dhaka, Bangladesh
1801042@iot.bdu.ac.bd

Nurjahan
Computer Science
Louisiana State University and Agricultural and Mechanical College
Baton Rouge, Louisiana, USA
nurja1@lsu.edu

Md Whaiduzzaman
Torrens University, Design and Creative Technology Vertical
Brisbane, Australia
md.whaiduzzaman@torrens.edu.au

## Abstract

The novel area of the Internet of Things (IoT) and robotic automation has experienced steady growth and massive breakthroughs in different industries including the hospitality industry. However, integrating these technologies into human-centric operations like restaurants presents unique challenges due to the requirement of complex interactions. This study addresses these issues by developing a tailored, cost-effective, self-sufficient robotic system for such settings. Our design includes an advanced path-following mechanism using black line tracing and supports multimodal interactions such as speech, text, and smartphone navigation. It also features sophisticated obstacle recognition and dynamic feedback capabilities to alert nearby individuals to potential obstructions, enhancing safety and efficiency. Our developed robot has sensors for precise food item retrieval and a digital interface for easy meal ordering via QR codes, integrated with an online payment system.

## CCS Concepts

• **Hardware → Wireless devices**; **Sensors and actuators**; • **Information systems →** *Web applications*; • **Computer systems organization → Robotic components**.

## Keywords

Internet of Things (IoT), multimodal interactions, self-sufficient robotic system, smartphone-based navigation, Obstacle Detection.

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## 1 Introduction

The dine-in sector is experiencing a digital transformation as cutting-edge technologies influence how patrons engage with eateries and oversee day-to-day activities. In countries with advanced technology such as Japan, South Korea, and Singapore, individuals are more frequently collaborating with robots to adjust to automation and reduce the likelihood of joblessness [25]. Apart from that, during the COVID-19 pandemic, people needed to maintain a safe distance from each other to stop the spread of the deadly virus, and it was dangerous to go near customers. During those challenging times, the number of restaurant staff gradually declined. In 2017, approximately 109,543 service robot units were sold globally, according to the International Federation of Robotics (2018), representing an 85% increase over 2016[4]. The global robotics industry is expected to reach USD 147.26 billion by 2025, with a CAGR of 17.24% between 2017 and 2025[6]. This paper proposes a new system that enhances the dining experience through the use of technologies such as QR code scanning, online ordering, and autonomous service delivery. Here, information can be found about the study's plan and execution, as well as the benefits concerning captured, organized, and analyzed data looking at the convenient processes for customers and employees of restaurants. By placing the QR codes on tables, customers can go to the restaurant's website and log into the restaurant to further customize their dining experience. The proposed system has included facilities for online payments, which will allow a hassle-free and safe transaction for the customers.Every detail about the order is communicated instantly to the chef's website, which is built into the hardware prototype. This integration also

enables the chef to have a daily order list of customers to help manage the customer's orders. Finally, once the food has been placed on the robot through the chef's preparation, the robot is operated via a control panel. For instance, the robot does not require human help to get to the table of the customers' choice, thus being able to deliver the orders to the right table. With regard to mobility, if the robot is fitted with features that include obstacle detection and avoidance, the robot is able to move around the restaurant without an increased likelihood of occurrence of accidents. Once the robot has guided the customer to the designated table, it waits for the customer to pick up food. Our experimental prototype has undergone several tests, demonstrating seamless operation and a positive user experience. Key features, such as voice alerts, obstacle detection, and avoidance, ensure safety and enhance user interaction. Our robots are designed to adapt to various demographic structures within different restaurant settings. By customizing the path lines and modifying the algorithm to suit specific layouts, we maintain the core algorithm's integrity while achieving optimal performance across diverse environments. This adaptability underscores the system's versatility and reliability, making it suitable for various applications. The system employs a backend management system that operationalizes the current architecture of a restaurant business. It is this system that allows the robot, kitchen staff, the chef, or whatever operation a restaurant has to be in harmony. The paper offers an analysis of the use and consequences of robotic systems in restaurants. Through the application of the intelligent and self-sufficient robots, this system can transform the conventional service manner in the restaurant business and decrease the reliance on manual labor, increase effectiveness, and increase customers' satisfaction. The primary contributions of this study are as follows:

I. Design and develop an IoT-enabled line-following food-serving robot for restaurants.
II. Develop an efficient web-based food ordering system for customers by scanning QR codes.
III. Integration of a mobile payment system using the web interface.
IV. An obstacle detection and avoidance mechanism is incorporated into the robot, and a voice command is provided to move the obstacle.
V. Implemented a system that automatically determines the empty tray after customers receive food and autonomously return to the kitchen.
VI. Our restaurant robot can accept commands through voice and touch, allowing for multi-modal operation.

The novelty of our work includes the following points:

I. We have developed a robust algorithm for line-following food-serving waiter robots, which is flexible enough to be modified for any type of restaurant without changing the core system.
II. Our system works with a multi-path mechanism, whereas most existing systems follow circular guided path navigation.

The remainder of this paper is organized as follows: Section 2 describes recent related work. Section 3 provides an in-depth analysis of the proposed approach and methodology. Section 4 presents the experimental data and analyses. Section 5 presents the paper's discussion, in which we contrast our work with some recent state-of-the-art works, cost analysis, and Coordination and Managerial Aspects of the Work. Finally, section 6 presents the conclusions of this work.

## 2 LITERATURE REVIEW

Numerous researchers have proposed IoT-based restaurant robots to automate food serving tasks. Hamdany et al. [8] proposed an interface between a customer and a waiter robot through an E-menu in their paper. The robot can take orders and serve the food on its respective table. The system also has an integrated Credit Card device for payments. Wan et al.[24] developed kinematic solutions. The robot can move at a higher velocity without crashing, spilling, or docking issues and ensure smooth velocity control. Consequently, the robot can easily serve liquid food and drinks without any problems. Ivanov and Webster[11] investigate public perceptions of using robots in food and beverage operations by analyzing survey data from 1,579 respondents across 12 languages. It examines how positive attitudes toward robots in tourism and hospitality relate to their acceptance in food and beverage settings, focusing on reliability, functionality, and advantages over human employees. Jang and Lee [12] examine how attributes of serving robots (anthropomorphism, animacy, likeability, intelligence, and safety) affect restaurant customers' perceptions and behaviors through an online survey of 294 respondents. It identifies significant relationships between these attributes, perceived benefits and risks, perceived value, satisfaction, and customers' intention to revisit, providing insights for sustainable restaurant development amid labor shortages. Seyitoğlu and Ivanov [21] investigate travelers' experiences at robotic restaurants worldwide, using reviews to gather qualitative data. It identifies six key components of the robotic restaurant experience: attraction for kids, robotic system, memorable experience, ambiance, food attributes, and deficiencies. Naidu and Rao [18] detail the development of a restaurant serving robot that uses remote navigation, a USB cam and Kinect V2 sensor for object identification, and a SLAM algorithm for table positioning. It features ultrasonic sensors for obstacle detection, a vacuum cleaner for floor cleaning, and a 3-DOF robotic arm with a soft gripper, simulated using Robo Analyzer and MATLAB, to interact with customers. El-Said and Al Hajri [7] examine how the COVID-19 pandemic has influenced consumer attitudes towards robot service restaurants in Muscat, focusing on factors affecting experience satisfaction and extension. It finds that perceived usefulness, speed of service, and novelty impact satisfaction, while perceived enjoyment and satisfaction influence the intention to extend the experience, with novelty seeking, risk reduction, and trust moderating these effects.

Akhund et al.[2] leveraged a waiter robot that could follow lines and avoid obstacles. They can also take and serve orders for customers. The robot reads the RFID from tables to recognize particular orders from customers. The collected data are stored in a cloud database. For this, the orders taken by the robots automatically go to the chefs. Moreover, they included an auto bill generation facility attached to the robot. Mishra et al. [17] developed a system in which customers can easily order food through an app via smartphones; it only needs to know the table number and the food that the customer orders. The restaurant's authority serves the food through the robot. Customers can add food to their menus. Through the

menu management system, the admin can add, delete, or edit food to the menu card. Kamruzzaman and Tareq in [14] investigated a waiter robot that was designed and implemented to receive order requests via Android apps, collect food or drinks from the kitchen boy, travel to a particular destination, and return to its source point after completing the order of requested items. Hasib et al.[10] designed and implemented a low-cost Arduino-based Android-application-controlled waiter robot. The robot can navigate, avoid obstacles, serve meals to specific customers, and autonomously return to the kitchen. It uses the PID algorithm for precise path following and a sonar sensor to detect obstacles and display messages and warnings on an LCD screen. Reddy et al. [20]implemented a web-based restaurant system. A dynamic database utility system was designed to retrieve information from a centralized database, prioritize user convenience, and aim for efficiency and accuracy to minimize human error. This system successfully overcame the shortcomings observed in previous similar systems and proved to be cost-effective in both development and operation. Malik et al. [15] focused on applying a serving robot, specifically an autonomous serving robot designed to take orders and serve food to customers. The prototype was developed using the available resources to minimize costs. This study highlights the concept of an automatic menu-serving robot that provides efficient service to restaurant customers. To place an order, customers can call the robot by pressing a switch on the table. The entire system operates using the RF technology. Prema [19] built an automated menu ordering system for hotels and restaurants. It replaces manual order entries with an electronic menu and wireless communication. Customers use electronic menu cards with a keypad to select items that are wirelessly transmitted to the kitchen. The system reduces delays and operates within a 30-meter indoor range using a microcontroller, display, and wireless transceiver modules. Shiny et al. [22] developed a web-based application for robotic restaurants in India to automate ordering systems. Compared with the current service robots in China and Japan, this system solves the problem of ordering robots. Customers can employ the smartphone application; it is convenient to use it. Restaurant and hotel chains enhance the performance of their stars and reduce the possibilities of human errors in order taking. This application aims for use in automating robotic restaurants, and it is somewhat different from online registration application that hotel business entities employ in the management of their clients and experimentation.

## 3 METHOD

In this section, we have discussed the project's features, algorithm, physical and web design, system model, and circuit diagram.

### 3.1 Working Procedure

The conceptual flow of the proposed system is shown in Figure 1. First, customers must try to connect to the restaurant's Wi-Fi. After a successful connection, customers need to scan the QR codes of their respective tables. The QR code redirects the customer to the order placement website. On the website, they must select their seat number, table, and food item according to their desired quantity. After the selection, the mobile payment interface opens. The payment was performed using the brash gateway. The order-related information is then saved into the database. After receiving an order, the chef prepares the food and places it in the robot's tray. Then the robot navigates to the specified customer.

### 3.2 Algorithm

The algorithm for reaching the destination after receiving a command from the chef and returning to the kitchen after receiving the food from the customer has been demonstrated in Algorithms 1 and 3, respectively. When the chef places food in the robot's tray and gives commands, the robot starts to move. In the beginning, it exits the kitchen checkpoint by rotating backward. It rotates backward in the right direction until the right IR detects the black line. After detecting the black line, the robot moved forward, following the LFR mechanism until a checkpoint was detected. We provide the LFR mechanism with obstacle detection. If our robot detects an obstacle using the front IR sensor, it will stop moving and emit a warning through voice until the obstacle moves away from its path. If the obstacle does not move after 5 seconds of the alert, the robot will use the obstacle avoidance algorithm 5 to navigate around the obstacle. We have considered two checkpoints for our project. At the first checkpoint, the robot checks both table numbers of the checkpoint. If the desired table number is not found here, it exits the first checkpoint and moves forward following the LFR mechanism until the 2nd checkpoint is detected. When the commanded table number is found, the robot rotates toward the matched table path. If the table number is 1 or 3, then it rotates in the right direction until the right IR detects a black line. If the table number is 2 or 4, it rotates in the left direction until the left IR detects a black line. Subsequently, the robot followed the LFR mechanism until it reached the table's checkpoint. After reaching the table's checkpoint, it stops its movement and gives a voice message to the customer once they receive the food and wait until the food tray is empty. After receiving the food from the customer, the IR on the tray detects that the tray is empty, and then the robot exits the table checkpoint by rotating backward. It will rotate in the right direction until the right IR can detect the black line. Then, it moves forward following the LFR mechanism until a checkpoint is detected. After getting to the checkpoint, it rotates itself toward the path of the kitchen checkpoint. If the robot comes from table no 1 or 3, then it rotates itself in the left direction; otherwise, it rotates itself in the right direction. Again, it moves forward following the LFR mechanism until the next checkpoint is detected. After getting one more checkpoint, it decides. If the robot comes from table no 1 or 2, then it considers the detected checkpoint as a kitchen checkpoint and ends its cycle. Otherwise, it exits the checkpoint by moving forward and continues following the LFR mechanism until the kitchen checkpoint is detected. After getting to the kitchen checkpoint, it stops movement and ends its cycle. The line following procedure and the rotation procedure of the robot are shown in the algorithm 3 and 4.

Our restaurant robot can navigate through crowds and avoid obstacles, even when multiple robots are present, using its obstacle detection and avoidance techniques as outlined in the algorithm 5

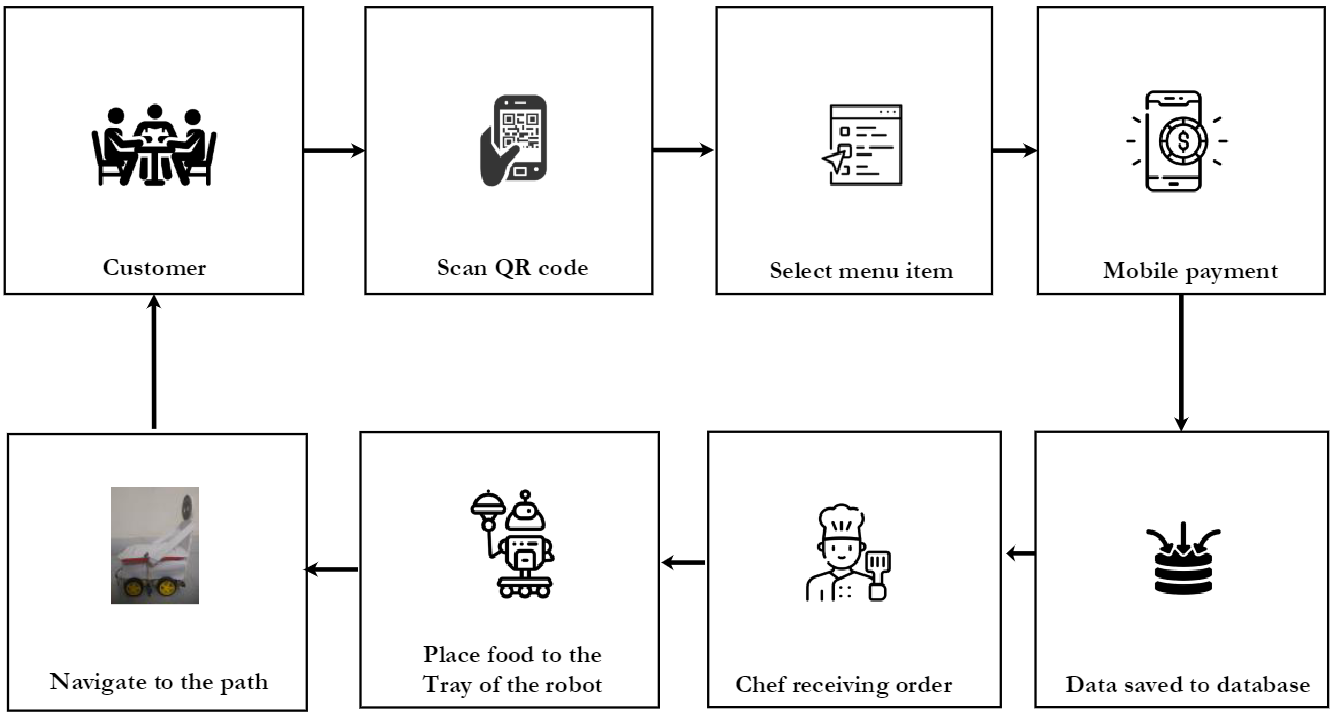



**Figure 1: Conceptual flow of the proposed system**

**Algorithm 1** Procedure to go from kitchen to table

```
Place food on the tray.
if destination value read from waiter then
    Make the right rotation before starting.
end if
if destination is table 1 then
    Go to 1st checkpoint and make the left rotation.
else if destination is table 2 then
    Go to 1st checkpoint and make the right rotation.
else if destination is table 3 then
    Go to the 2nd checkpoint and make the left rotation.
else if destination is table 4 then
    Go to the 2nd checkpoint and make the right rotation.
end if
After rotation, go to the table's stop point.
At the stop point, give a voice message.
```

**Algorithm 2** Procedure to return to kitchen from table

```
if tray is empty then
    Make the right rotation before starting.
end if
if destination is table 1 or 3 then
    Return to the checkpoint and make the right rotation.
else if destination is table 2 or 4 then
    Return to checkpoint and make the left rotation.
end if
After rotation, return to the kitchen's stop point.
```

**Algorithm 3** Line following procedure

```
if both left and right IR sensors detect white surface then
    Move forward.
else if both left and right IR sensors detect black surface then
    Stop moving.
else if left IR sensor detects black surface then
    Move left.
else if right IR sensor detects black surface then
    Move right.
end if
if any object is detected in front then
    Stop moving and warn by voice message.
end if
```

**Algorithm 4** Rotation procedure

```
while rotation is triggered do
    if rotating left then
        Rotate left until the left IR sensor detects a black surface.
    else if rotating right then
        Rotate right until the right IR sensor detects a black surface.
    end if
end while
```

**Algorithm 5** Obstacle Avoidance in Crowds

```
if any obstacle is detected in front then
    Stop moving and warn by voice message.
end if
if waiting time is greater than 5 seconds then
    Start to avoid the obstacle.
end if
Rotate to the left and check if the path is clear.
if path is clear then
    Move forward for 2 seconds and rotate to the right.
else
    Check if the right path is clear.
    if path is clear then
        Move forward for 2 seconds and rotate to the left.
    end if
end if
After rotation, move forward for 2 seconds and try to return to the line by moving left or right.
```

## 3.3 System Arrangement

RoboServe's system configuration is intended to assist with effective order management, food preparation, and delivery inside the restaurant environment. The system's primary components include four tables, a kitchen, and two checkpoints that are carefully placed to maximize the robot's movement. Each table was positioned to maximize space use and ease of navigation. The table was set up as follows:

- Table 1: Positioned on the left side of Checkpoint 1
- Table 2: Located on the right side of Checkpoint 1
- Table 3: Situated on the left side of Checkpoint 2
- Table 4: Positioned on the right side of Checkpoint 2

The system has two checkpoints to ensure smooth navigation and rapid delivery. The checkpoints were placed throughout the restaurant to maximize the robot's progress. Checkpoint 1 acts as the robot's initial reference point for its location and orientation within the restaurant, which is located near the kitchen area, to allow the robot to pick up food more easily. Checkpoint 2 also references the rest of the tables (Table 3 and Table 4). By tailoring the path lines and adjusting the algorithm to accommodate specific layouts, we preserve the core algorithm's integrity while ensuring optimal performance in various environments.

## 3.4 System Design

The system design incorporated a number of IR sensors along with microcontroller, switches, controllers, actuators, batteries and other hardware and mechanical parts. A significant contribution of the ESP32 microcontroller was observed by the researchers. The robots and was used for the processing activities. This microcontroller contributed to data transfer after which information security was enhanced and effective transmission of customer orders as well as invoices to the chef's website. IR sensors are used to detect and measure infrared radiation. The robot follows the line with the help of an IR sensor. The sensors help the robot detect obstacles by emitting IR light and measuring reflected or blocked signals. It also detects whether the customer receives food from the robot's tray. Direct current (DC) motors were used to drive the movement of the robot. The motor shaft that was connected at the wheels allowed the robot to move in the correct formation. For motor controlling and regulation, a motor driver, L298N, was employed to drive the motors. These were used to control the robot's movement through center and right buttons. They guarantee free and accurate motor control by regulating aspects like speed, direction, throttle, gear shift, braking and torque. Also, we redesigned mini breadboards for further exploitation and pressed a metal strip for switching between midi and mini breadboards.to build and troubleshoot electronic circuits, therefore preventing any requirement solder. Moreover, a four-wheel drive chassis is intended to assist and regulate a car with power delivered to all four wheels. Using a 4WD chassis in the robot improves mobility, agility, and load-carrying capabilities, resulting in improved performance in restaurant settings. A Bluetooth speaker mainly listens to the robot's response loudly. The voice passes through the system's app to the Bluetooth speaker. There is no direct connection between the robot and the speaker. However, the voice text is written in the code. Finally, we used 3.7V 8000mAh Ultra fire batteries to power the robot. The block diagram of the proposed system is shown in Figure 2. The overall system

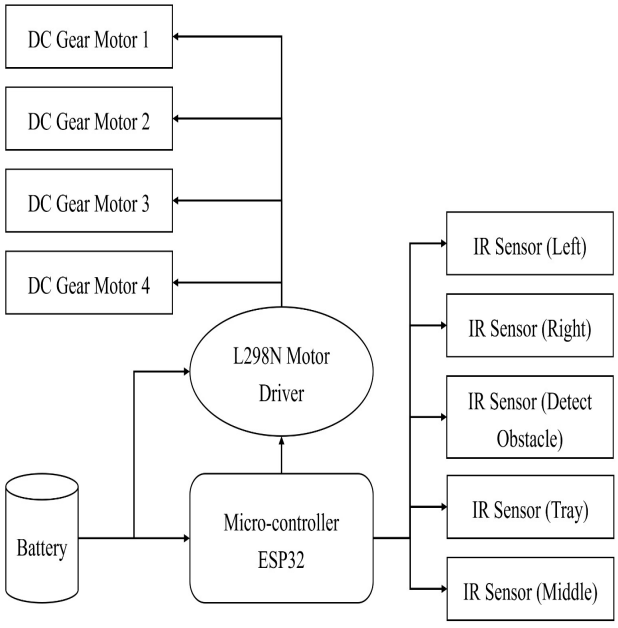



**Figure 2: Block Diagram of the Robotic System**

architecture flow is shown in Figure 3.

## 3.5 Web Interface Design & Functionalities

Each table in the restaurant has an attached QR code. The user-friendly restaurant website was opened after scanning. Then, by entering the specified table and seat numbers, the restaurant menu is opened. The total payable bill was shown after the food selection from the menu. Once payment is complete, an order is marked complete and moved to the next process area, where it is

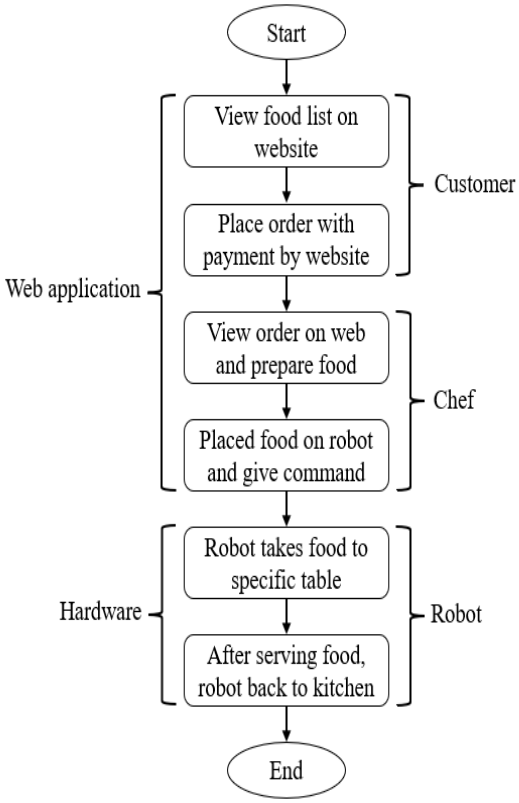



**Figure 3: System Architecture**

displayed on the chef or management employee's dashboard, and they may process it. Here we have incorporated the Bkash payment-getting system, which is the most popular online banking system in Bangladesh, to make transactions safer and more convenient. The order processing and transaction system interface is shown in Figure 4. The front end of the web interface is designed with HTML, CSS, and JavaScript, and PHP and MySQL databases are used for the back end.

The overall food order and serving system flow is shown in Figure 5.

# 4 RESULTS

## 4.1 Experimental Prototype

The circuit diagram of the experimental prototype is shown in Figure 6. To enable the utilization of our restaurant robot, a black line must be incorporated alongside the tables within the establishment. The robot is equipped with three IR sensors that detect the presence of the black line. An IR module emits an infrared ray and receives its reflection to identify the black line. Subsequently, the data is transmitted to the esp32, which processes the signals using our predetermined algorithm. The esp32 then sends the data to the motor driver, initiating the operation of the DC motor. As a result, the robot follows the line and moves accordingly. We developed the prototype shown in Figure 7. It is wrapped in paper and made of wood. This experimental prototype, designed for research purposes, can handle a payload of up to 1.50 kg. If the weight exceeds this limit, performance may degrade. It is not intended to be an operational prototype; however, the methods employed can be utilized to develop restaurant robots tailored to specific requirements.

## 4.2 Result Analysis

In this section, we analyze the system's results based on our experimental the outcomes. Focusing on several test points, the success rates of the robot's performance were evaluated based on the specific tasks. A summary of the results is as follows.

*4.2.1 Black line following.* After 100 attempts, the robot successfully followed the right line 94 times, resulting in a high success rate. of 94%.

*4.2.2 Obstacle Detection.* When faced with obstacles, the robot stops and gives voice notes to move on its path. After 100 attempts, obstacles were successfully detected 95 times, achieving a success rate of 95%.

*4.2.3 Food delivery on the specified table.* After 100 attempts, the robot successfully delivered food 97 times, thereby achieving a success rate of 97%.

*4.2.4 Empty tray detection.* : We tested this feature 100 times, and it successfully determined the empty tray in 97%, eventually obtaining a 97% accuracy rate.

*4.2.5 Ordering.* After 100 attempts, the robot took orders 99 times, resulting in a success rate of 99%.

*4.2.6 Data sending.* After 100 attempts, the robot successfully sent the collected data to the designated recipients (database, manager, and chef) 100 times, thereby achieving a success rate of 100%.

*4.2.7 Bill processing.* After 100 attempts, the robot accurately generated bills with the correct amounts 98 times, with a success rate of 98%.

# 5 DISCUSSIONS

The main objective of this study is to design a robot that is capable of serving food to customers at a specified table in a restaurant. In this section, We compare our proposed method with some of the state-of-the-art methods presented in Table 1. In the works of [1–3, 5, 8, 9, 13–15, 23], as per our knowledge they did not utilize a complete order-taking system using QR code, mobile payment system through a web application, and a robot controlling mechanism system through voice and manual, both at the same time. Hamdany et al.[8] implemented line-following and individual table identification features. In contrast, they did not employ several features that we have implemented, such as obstacle detection, order-taking procedure, payment procedure, and controlling mechanism features. Akhund et al. [2] designed a low-cost line follower robot with an obstacle detection facility, and it is also capable of identifying individual tables; however, it lacks other features such as online order taking, mobile payment systems, manual and voice control facilities, and empty tray detection. Kamruzzaman and Tareq [14] exploited the wheel-step counting technique to navigate to the destination instead of the following line. Features such as obstacle detection,

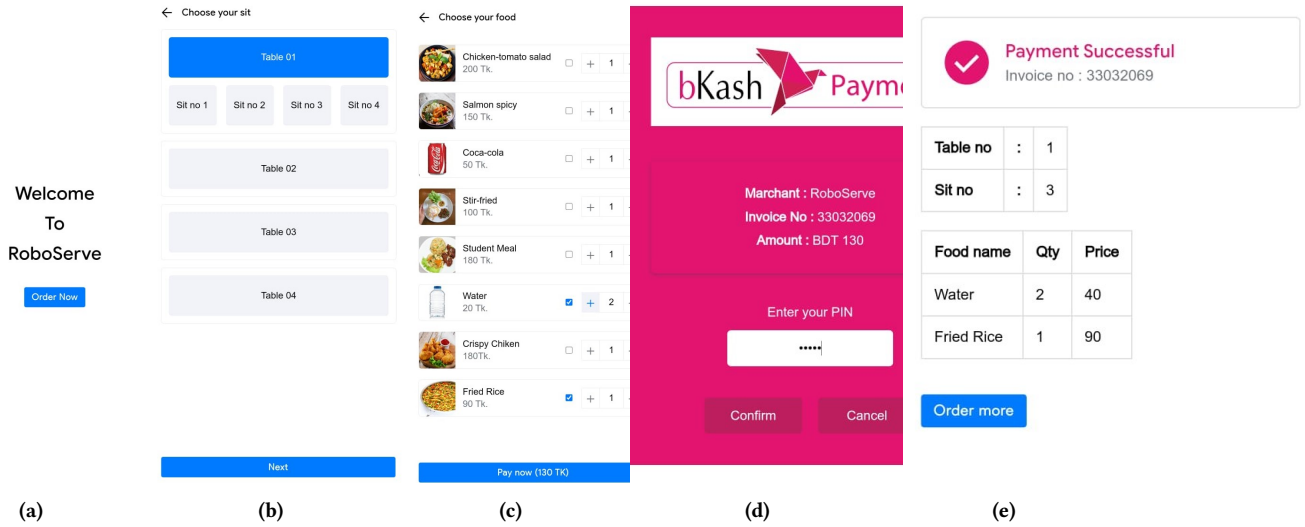



**Figure 4: Web Interface of Order Processing & Payment System (a) Home Page (b) Table and seat number page (c) Menu and the payable amount (d) Bkash Payment Process (e) Invoice generation**

| Reference | Line Following Capability | Obstacle Detection | Online Order Taking Through QR Code | Mobile Payment Through Web App | Manual & Voice Controlling Facility | Individual Table Detection | Empty Tray Detection | Cost Effective |
|---|---|---|---|---|---|---|---|---|
| [10] | Yes | Yes | Yes | Yes | Yes | No | No | Yes |
| [8] | Yes | No | No | No | No | Yes | No | No |
| [2] | Yes | Yes | No | No | No | Yes | No | Yes |
| [21] | No | No | No | No | No | Yes | No | Yes |
| [14] | No | No | No | No | No | Yes | Yes | Yes |
| [13] | No | Yes | No | No | No | Yes | No | Yes |
| [3] | Yes | No | No | No | No | Yes | No | Yes |
| [23] | Yes | Yes | No | No | No | Yes | No | No |
| [9] | Yes | Yes | No | No | No | No | No | Yes |
| [5] | Yes | Yes | No | No | No | No | No | No |
| [1] | Yes | Yes | No | No | No | Yes | No | No |
| [16] | Yes | Yes | No | No | No | Yes | No | Yes |
| **Proposed work** | **Yes** | **Yes** | **Yes** | **Yes** | **Yes** | **Yes** | **Yes** | **Yes** |

**Table 1: Comparison with some of the state-of-the-art works**

payment facilities, and voice, as well as manual control facilities, are absent in their system. Jyh-Hwa and Kuo[13] implemented a fixed-path guided system. They used a laser-positioning system for rapid and precise positioning. The system can identify obstacles and individual tables, and the robot can show customers the unoccupied table. However, the system lacks features like online order taking through a QR code, mobile payment through a web application, manual and voice controlling facility, and empty tray detection. Asif et al [3] implemented line following and individual table identification features. However, they have not employed several features, such as obstacle detection, order-taking procedure, payment procedure, and controlling mechanism features. Thanh et al. [23] worked on a line-following robot that can detect obstacles and identify individual tables. However, the system is not similar to the other features of our proposed work, such as the process of online orders, payments, and control facilities. Hasan et al. [9] designed a low-cost line follower robot in which every line has different colors. The robot can differentiate among various colors and choose the desired one to find its target. The work also offers an obstacle detection facility, but it lacks other features, such as online order taking through QR codes, mobile payment systems, manual and voice control facilities, and empty tray detection. Bhat

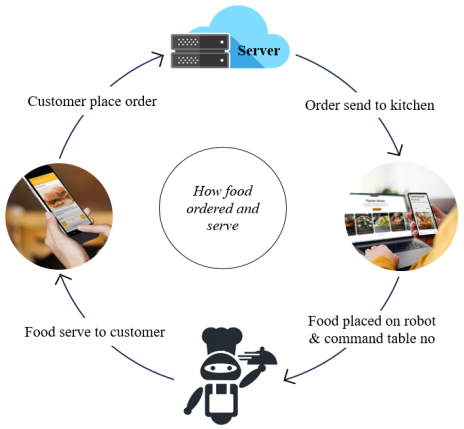



**Figure 5: Food Order and Serving System**

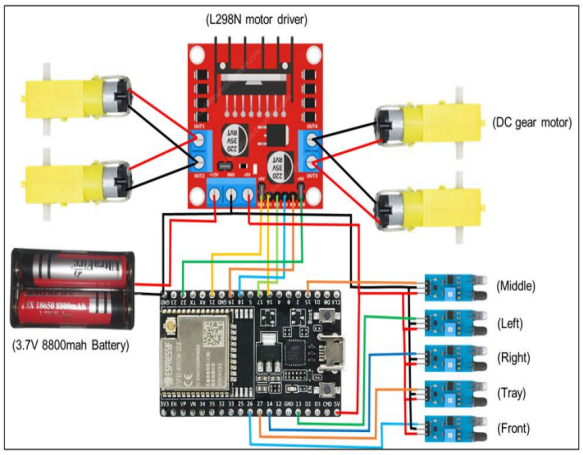



**Figure 6: Circuit Diagram**

and Meenakshi [5] followed the line-following mechanism for their system. This system can also detect obstacles. However, the system cannot properly fulfill the features listed in Table 1. Afsheen et al. [1] designed a line follower robot with an obstacle detection facility and identified individual tables for serving orders; however, it did not employ other features such as online order taking through QR codes, mobile payment systems, manual and voice control facilities, and empty tray detection. Malik, Singh, et al.[16] developed a low-cost line-follower robot with obstacle detection capability and the ability to identify particular tables. However, it lacks additional capabilities, such as online order-taking procedures, mobile payment systems, manual and voice control facilities, and empty tray detection. Our proposed method is based on a line-following robot that can detect obstacles and identify individual tables for serving orders. In our system, customers can order food by scanning QR codes, paying the bill through a web application, or using manual touch or voice control facilities from our developed website; the chef or manager can command a serving robot, and an automated detection system is employed to check whether the customer receives the food from the tray or not. In addition, we designed the entire system at the expense of a meager low cost. The cost of the prototype was approximately five thousand six hundred BDT. With reference to the issues highlighted above, one can probably conclude that clear that the proposed system provides superior performance as compared to the methods. Whereas more sophisticated bot servants are capable of delivering food without it. In the following lines, they employ very sophisticated technology, make them expensive and complex. In contrast, our proposed system brings a cost-effective and efficient way of running a restaurant. The robot that engages in operations in a manner that employs line-following technologies but does not depend on them, on advanced mechanics.

Still, the proposed method brings many advantages in the evaluation for the influence of the node and edge attributes of complex networks and real-world scenarios. However, there are also few possible limitations for their usage during work-days. With this, though it can point out objects that are within its recognition range and give directions on the course to follow. To avoid them and to shift things around, there has to be a black line or path to direct the robot. Without a path, a robot can not reach to the destination table of the restaurant. Moreover, it can take single order at a time, and After all, it is quicker to deliver only one order at a time. Another important concern is performance

## 5.1 Cost Analysis

In a typical restaurant in Bangladesh, a waiter's salary is approximately 15,000 BDT per month, amounting to 180,000 BDT per year. Given that a single waiter is often insufficient for optimal service, our tests indicate that RoboServe can effectively handle the workload of two waiters, thus saving the salary costs of two employees. Consequently, the annual cost for two waiters totals 360,000 BDT. In contrast, the fully developed operational RoboServe robot costs around 50,000 BDT, with an additional maintenance cost of 10,000 BDT per year. Therefore, the annual savings for each restaurant deploying RoboServe is shown in the calculation 1.

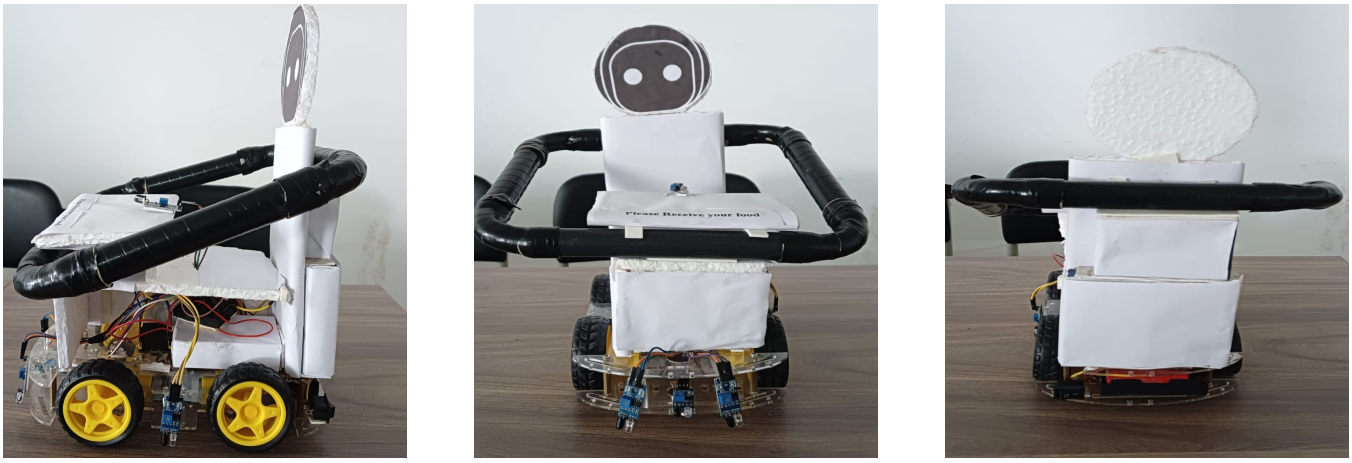
$$\begin{aligned} \text{Annual Savings} &= 360{,}000\,\text{BDT} - (50{,}000\,\text{BDT} + 10{,}000\,\text{BDT}) \\ &= 300{,}000\,\text{BDT Approx.} \end{aligned} \tag{1}$$

As stated, the system is quite cost-effective, designed using common hardware, and implemented in the simplest manner possible compared to existing systems. This ensures that the operational system will be compatible with all types of restaurants. Although we have worked with an experimental prototype, the cost analysis of this prototype is provided in table 2.

Since this is still a functional prototype, a durability test has not been conducted. Once a deployment prototype is developed, a durability test must be performed. Implementing robots like RoboServe does not exacerbate unemployment issues; instead, it creates opportunities for higher-level jobs in robot manufacturing and maintenance.

## 5.2 Coordination and Managerial Aspects of the Work

To address the applicability of the proposed robotic system in restaurant environments, technical issues are not the only focus of our




(a) (b) (c)

**Figure 7: Experimental Prototype (a) Side View; (b) Front Side View; (c) Back Side View;**

**Table 2: Total Cost of the Prototype**

| Hardware Items | No of Item | Cost (BDT) |
|---|---|---|
| ESP32 | 1 | 500 |
| IR sensor | 3 | 210 |
| Motor driver L298N | 1 | 200 |
| 4WD car chassis kit | 1 | 1200 |
| Ultrasonic sensor | 1 | 90 |
| Jumping wire | - | 200 |
| Speaker | 1 | 500 |
| Robot body | 1 | 500 |
| Battery | 2 | 200 |
| VPS for host website (per month) | - | 2000 |
| **Total Cost** | | **5600** |

study; managerial factors and coordination mechanisms, as well as safety aspects, are also considered. The system employs a highly developed backend management system that conveniently fits into the restaurant facilities: the robots, kitchen workers, chefs, and other operational employees. This integration helps to provide a live view of the customer's order and tackles issues to do with the flow of communication between them and other customers in the proper way as designed. In the same respect, the design of the system incorporated safety features of detecting and avoiding static and moving obstacles in addition to providing dynamic feedback to the people within the environment about the presence of obstructions within the vicinity. This points out that with the managerial and safety aspects put forward in the proposed solution, there is the needed approach to improve efficiency and customer satisfaction in the restaurant industry.

## 6 CONCLUSION

The idea of using a robot to deliver food is not new; however, there are several technological challenges to solve, and the proposed method offers significant advantages in terms of convenience, efficiency, and customer satisfaction. It successfully demonstrated the seamless integration of technology into the dining experience, allowing customers to easily access the menu, place orders, and make secure payments through a user-friendly website. The successful outcome of this project was the discovery that the application of technology in the restaurant business may greatly enhance all functional aspects of the restaurant. All workflows for placing an order, payment for food consumption, and delivery of the foods are made through an automated system. These reduce wait times and mistakes; here is an instance of how technology can change the restaurant sector, focusing on customers and making the existing experiences as pleasant and convenient as possible. It is possible that this work serves as a model for integrating robotic operations into smart restaurants. Adopting innovation and capturing the capabilities of the robotic prototypes gives restaurants an edge in the market, as well as offers more impressive services to customers.

## References

[1] Salmiya Afsheen, Lubna Fathima, Zabiha Khan, and Mohammed Elahi. 2018. A self-sufficient waiter robo for serving in restaurants. *International Journal for Advance Research and Development* 3, 5 (2018), 57–67.

[2] Tajim Md Niamat Ullah Akhund, Md Abu Bakkar Siddik, Md Rakib Hossain, Md Mazedur Rahman, Nishat Tasnim Newaz, and Mohd Saifuzzaman. 2020. IoT Waiter Bot: a low cost IoT based multi functioned robot for restaurants. In *2020 8th international conference on reliability, infocom technologies and optimization (Trends and Future Directions)(ICRITO)*. IEEE, 1174–1178.

[3] M Asif, M Sabeel, and KZ Mujeeb-ur Rahman. 2015. Waiter robot-solution to restaurant automation. In *Proceedings of the 1st student multi disciplinary research conference (MDSRC), At Wah, Pakistan*. 14–15.

[4] Katerina Berezina, Olena Ciftci, and Cihan Cobanoglu. 2019. Robots, artificial intelligence, and service automation in restaurants. In *Robots, artificial intelligence, and service automation in travel, tourism and hospitality*. Emerald Publishing Limited, 185–219.

[5] Sandeep Bhat and M Meenakshi. 2015. Embedded System based waiter and military robot path planning. In *2015 International Conference on Control, Instrumentation, Communication and Computational Technologies (ICCICCT)*. IEEE, 507–510.

[6] A Dabral, Akash Rawat, Siddhartha Juyal, Satish Joshi, and Sumit Pratap. 2022. Robotic in modern-day restaurants and its impact on the dining experience. *International Journal of Mechanical Engineering* 7 (2022), 139–148.

[7] Osman El-Said and Sara Al Hajri. 2022. Are customers happy with robot service? Investigating satisfaction with robot service restaurants during the COVID-19 pandemic. *Heliyon* 8, 3 (2022).
[8] Arwa Hamid Salih Hamdany, Lubab H Albak, and Raid Rafi Omar Al-Nima. 2019. Wireless waiter robot. *TEST Engineering & Management, The Mattingley Publishing Co., Inc* 81 (2019), 2486–2494.
[9] Kazi Mahmud Hasan, Abdullah Al-Nahid, KJ Reza, S Khatun, and MR Basar. 2013. Sensor based autonomous color line follower robot with obstacle avoidance. In *2013 IEEE Business Engineering and Industrial Applications Colloquium (BEIAC)*. IEEE, 598–603.
[10] Khan Md Hasib, Md Atiqur Rahman, Marzia Tahsin, Asif Karim, Sami Azam, and Friso DeBoer. 2022. LFR Waiter: Arduino Based Android Application Controlled Waiter Robot. In *TENCON 2022-2022 IEEE Region 10 Conference (TENCON)*. IEEE, 1–6.
[11] Stanislav Ivanov and Craig Webster. 2023. Restaurants and robots: public preferences for robot food and beverage services. *Journal of Tourism Futures* 9, 2 (2023), 229–239.
[12] Ha-Won Jang and Soo-Bum Lee. 2020. Serving robots: Management and applications for restaurant business sustainability. *Sustainability* 12, 10 (2020), 3998.
[13] Tzou Jyh-Hwa and L Su Kuo. 2008. The development of the restaurant service mobile robot with a laser positioning system. In *2008 27th Chinese Control Conference*. IEEE, 662–666.
[14] Md Kamruzzaman and Md Tareq. 2017. Design and implementation of a robotic technique based waiter. In *2017 3rd International Conference on Electrical Information and Communication Technology (EICT)*. IEEE, 1–5.
[15] Neeti Malik, Neetu Rani, Pratibha Alpana Singh, and Srishti Pragya. 2016. Review paper on-serving robot new generation electronic waiter. *IJIRST–International Journal for Innovative Research in Science & Technology* 2, 11 (2016), 775–777.
[16] Neeti Malik, Alpana Singh, Neetu Rani, and Poonam Pratibha. 2016. Serving robot: new generation electronic waiter. *International Journal of Engineering Science and Computing* 6, 4 (2016).
[17] Neelima Mishra, Dinesh Goyal, and Ashish Dutt Sharma. 2018. Automation in restaurants: ordering to robots in restaurant via smart ordering system. *Int J Technol Manage* 4, 1 (2018), 1–4.
[18] R Naidu and S Srinivasa Rao. 2023. Smart restaurant multi-purpose serving robot. In *AIP Conference Proceedings*, Vol. 2788. AIP Publishing.
[19] G Prema. 2017. Smart Menu Ordering System in Restaurant. *Int. J. Electron. Commun. Enginerring* 2017 (2017), 59–63.
[20] CAFK Sainath Reddy, KGK Chaitanya, M Abhinav, and KTH Feiroz. 2016. An online food court ordering system. *Journal of Information Technology & Software Engineering* 6, 4 (2016), 1–3.
[21] Faruk Seyitoğlu and Stanislav Ivanov. 2022. Understanding the robotic restaurant experience: a multiple case study. *Journal of Tourism Futures* 8, 1 (2022), 55–72.
[22] JS Shiny, AM Kumar, V Nanthagopal, and R Raguram. 2017. Automation of Restaurant (Ordering, Serving, Billing). *International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering* 6, 3 (2017), 1532–1539.
[23] Vo Nhu Thanh, Dang Phuoc Vinh, Ngo Thanh Nghi, et al. 2019. Restaurant serving robot with double line sensors following approach. In *2019 IEEE International Conference on Mechatronics and Automation (ICMA)*. IEEE, 235–239.
[24] Ash Yaw Sang Wan, Yi De Soong, Edwin Foo, Wai Leong Eugene Wong, and Wai Shing Michael Lau. 2020. Waiter robots conveying drinks. *Technologies* 8, 3 (2020), 44.
[25] Evanthia K Zervoudi. 2020. Fourth industrial revolution: opportunities, challenges, and proposed policies. *Industrial Robotics-New Paradigms* (2020).